

**Homogenizing Effect of Large Language Models (LLMs) on Creative Diversity: An Empirical Comparison
of Human and ChatGPT Writing**

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Abstract

Generative AI systems, especially Large Language Models (LLMs) such as ChatGPT, have recently emerged as significant contributors to creative processes. While LLMs can produce creative content that might be as good as or even better than human-created content, their widespread use risks reducing creative diversity across groups of people. In the present research, we aimed to quantify this homogenizing effect of LLMs on creative diversity, not only at the individual level but also at the collective level. Across three preregistered studies, we analyzed 2,200 college admissions essays. Using a novel measure—the diversity growth rate—we showed that each additional human-written essay contributed more new ideas than did each additional GPT-4 essay. Notably, this difference became more pronounced as more essays were included in the analysis and persisted despite efforts to enhance AI-generated content through both prompt and parameter modifications. Overall, our findings suggest that, despite their potential to enhance individual creativity, the widespread use of LLMs could diminish the collective diversity of creative ideas.

Keywords: Homogenization; collective diversity; diversity growth rate; generative AI; Large Language Models (LLMs).

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Creativity centers on generating new ideas (Green et al., 2024; Kaufman & Sternberg, 2019; Prabhakaran et al., 2014; Runco & Jaeger, 2012), driving individual problem-solving, and societal progress (Henriksen et al., 2016; Zwir et al., 2022). The recent advent of generative Artificial Intelligence (AI) systems has significantly changed how people perceive and engage in creative processes (Epstein et al., 2023; Franceschelli & Musolesi, 2025; Grilli & Pedota, 2024). These technologies are being used to assist human involvement in various creative areas once believed to be exclusive to human capabilities, including painting (e.g., DALL-E and Mid Journey), musical composition (e.g., Jukebox), and even video production (e.g., Sora). Among these generative AI systems, Large Language Models (LLMs), such as ChatGPT, stand out for their versatility in supporting various creative processes (Grilli & Pedota, 2024). While LLMs might be able to produce content that is as good as or even more creative than content generated by humans, their widespread use risks reducing the collective diversity of creative outputs (Bender et al., 2021; Castro et al., 2024). In the present research, we aimed to quantify this homogenizing effect of LLMs on collective diversity in creative outputs.

How do LLMs such as ChatGPT compare to humans as creative entities? Emerging research suggests that LLMs can not only match but also outperform people, at least on a few established measures of creativity. GPT models have been shown to outperform humans on the Alternative Uses Task (AUT; Guilford, 1967)—one of the most widely-used measures of creativity—although the most creative humans still outperformed GPT-3.5 and GPT-4 (Koivisto & Grassini, 2023). However, on the Torrance Tests of Creative Thinking (TTCT; Torrance, 1966), a widely used measure of divergent thinking, GPT-4 scored in the top 1% for originality and fluency, suggesting that in some cases, AI performance can reach the height of exceptionally creative humans (Guzik et al., 2023). Moreover, GPT-4 outperformed 85.3% of humans in making remote associations in the Divergent Association Task (DAT;

Olson et al., 2021) and exceeded 96% of humans when adopting the greedy search strategy (Chen & Ding, 2023). Beyond lab tasks, LLMs can generate creative outputs comparable to human-written texts, including creative ideas (Girotra et al., 2023; Meincke et al., 2024), stories (Orwig et al., 2024), poetry (Köbis & Mossink, 2021), motivational messages (Cox et al., 2023), and persuasive political messages (Bai et al., 2023; Hackenburg et al., 2023). Unsurprisingly, when humans and AI work together, the output is often more creative than when humans work alone. In a creative writing task, for example, a GPT-3 powered system helped untrained crowd workers to write better by improving grammar and introducing new ideas compared to writing without AI assistance (Lee et al., 2022). However, research also indicates that LLMs do not perform as well as humans in convergent thinking. For example, Claude 3.5 Sonnet showed worse performance in an abstract reasoning task (i.e., the *New York Times* Connections game) compared to both human novices and human experts (Samdarshi et al., 2024).

Paradox of AI-Driven Homogenization

Even in cases where LLMs can match human performance in creative tasks and promote the creativity of individuals, these AI systems may still hamper the collective diversity of human ideas. On the surface, it seems that enhancing the creativity of each person should also boost the creativity of the group. It is also possible, however, that the widespread use of AI may have a homogenizing effect on creative ideas, wherein the diversity of new ideas and perspectives diminishes across a group of people who use the same AI model (Anderson et al., 2024; Franceschelli & Musolesi, 2025; Padmakumar & He, 2024). This possibility is consistent with the idea of algorithmic monoculture, which suggests that the consistent use of a single prevailing AI system, such as ChatGPT, to conduct various tasks that used to be conducted by many different people can result in increased homogeneity of outcomes (Bommasani et al., 2022; Kleinberg & Raghavan, 2021). LLMs are designed to rely on probability distributions; consequently, the outputs of LLMs tend to lack surprise or novelty, as these models are fundamentally trained to conform to existing data distributions that are biased in favor of predominant cultural ideas

(Bender et al., 2021; Franceschelli & Musolesi, 2025). LLMs can thus exclude a variety of unique perspectives from different people. In short, using LLMs may lead to better outcomes for individual creators but at the cost of these outcomes becoming more similar across creators.

Several studies provide initial support for the paradox of AI-driven homogenization in the creative process. For example, compared to those who wrote alone, writers exposed to GPT-4 generated ideas produced better and more enjoyable stories, as evaluated by independent raters (Doshi & Hauser, 2024). These LLM-inspired stories were, however, more similar to each other than stories written by humans alone. Similarly, individuals using the fine-tuned InstructGPT model—which employed instruction tuning with Reinforcement Learning guided by Human Feedback (RLHF)—produced short essays that were more similar in terms of lexical and content diversity than those written without any LLM support (Padmakumar & He, 2024). In another study, people generated more ideas when using GPT-3.5 than when using the Oblique Strategies deck, a non-AI tool consisting of prompt cards designed to stimulate creative thinking (Anderson et al., 2024). However, ideas generated with GPT-3.5 were more semantically similar to each other than those produced with Oblique Strategies. Moreover, in a seven-day lab experiment followed by a 30-day follow-up, college students using ChatGPT, compared to those who did not, generated more creative responses to Alternative-Use-Tasks (AUTs) and problem-solving tasks as evaluated by independent raters (Liu et al., 2024). However, this enhancement came at the cost of homogenization across different individuals: ChatGPT-assisted content lacked the diversity of human-only responses. Furthermore, students who initially relied on ChatGPT continued to produce less diverse, AI-like content even in the absence of ChatGPT, while their individual creativity reverted to baseline levels.

Moreover, in a task that required creative problem-solving by finding innovative uses for everyday objects (i.e., MACGYVER), individual GPT-4 responses exhibited generally comparable performance to individual humans (Tian et al., 2024). However, at the group level, GPT-4's responses

were more homogeneous, limiting the benefits of aggregating multiple responses. In contrast, collective humans generated responses more diverse and varied more in accuracy across tasks, as individual differences in familiarity and competence contributed to a broader range of solutions. As a result, humans collectively outperformed GPT-4, underscoring the advantage of human collective creativity over AI-generated solutions. In sum, this literature lends initial support to the paradox of LLMs promoting individual creativity while simultaneously reducing diversity in creative outputs across different people.

The Present Research

In the present study, we develop a novel metric—the diversity growth rate—to quantify the homogenizing effect of LLMs. This metric dynamically captures the extent to which each additional creative output contributes to collective diversity, with a lower diversity growth rate indicating a stronger homogenizing effect. Going beyond prior approaches that primarily compared collective diversity statically between groups with a fixed number of outputs (Anderson et al., 2024; Cox et al., 2023; Doshi & Hauser, 2024; Liu et al., 2024), our diversity growth rate tracks how collective diversity evolves and saturates as more creative outputs are introduced. The present research thus provides a new tool to quantify AI homogenization with broad applications for research and practice. Across three preregistered studies, we analyzed 2,200 college admissions essays and compared the diversity of ideas in essays written by actual college applicants with those generated by GPT-4. We were thus able to examine the homogenizing effects of GPT-4 in the context of real-world creative outputs rather than a controlled lab task, showcasing its potential to quantify AI homogenization across a wide variety of domains.

In Study 1, we aimed to establish a baseline comparison between human-written and GPT-4 generated college admissions essays. In Study 2, we aimed to replicate the findings of Study 1 while examining whether the homogenizing effect of GPT-4 would persist after enhancing the creativity of the

output via prompt and parameter modifications. In Study 3, we aimed to replicate the findings in a much larger sample of 1,600 essays and to expand our findings by enhancing the AI output via Chain-of-Thought (CoT) prompt engineering (Meincke et al., 2024; Wei et al., 2023). We adopted the CoT prompting as a state-of-the-art approach to enhancing the creativity and diversity of GPT-4 essays. To contextualize the effects, we also examined whether diversifying the racial and ethnic backgrounds of human authors would increase the diversity growth rate.

Across all experiments, we quantified the diversity of ideas using semantic distance, which measures the degree of proximity between the meanings of words. Semantic distance has emerged as a valid and reliable method for assessing creativity by capturing the originality and novelty of semantic associations (Acar & Runco, 2019; Green, 2016; Hass, 2017; Heinen & Johnson, 2018; Johnson et al., 2022; Kenett, 2018, 2019; Mikolov et al., 2013). Specifically, semantic distance captures the integration of diverse and potentially novel concepts and ideas in written content, reflecting a greater level of the divergent thinking aspect of creativity (Cox et al., 2021, 2023; Johnson et al., 2022).

General Method

Ethical compliance

This research was approved by the institutional review board of Georgetown University (STUDY00004661) and complied with all relevant ethical regulations.

Preregistration

We preregistered all the studies, including predictions, study design, sample size, measures, and analysis plans, on the Open Science Framework (OSF) at <https://osf.io/ywap5/registrations> (see also Table S8). Study 1 was registered on July 5, 2023; Study 2, on August 24, 2023; and Study 3, on April 8, 2024. During the analysis, we made several deviations from our original protocol for Studies 1 and 2, which were necessary to address unforeseen issues and to enhance the robustness of our findings.

Following the best practices (Willroth & Atherton, 2024), we documented all deviations and unregistered steps, including their reasons and expected impacts (see Tables S9–S10).

Essay Data

Human-Written Essays

We randomly selected human-written essays used in Studies 1–3 from the archival data submitted by applicants to a private East Coast university from 2018 to 2022 before the public launch of ChatGPT. The dataset consists of 49,804 essays after excluding 1,000 essays used in our pilot studies. The applicants wrote their essays in response to the following prompt: *“As Georgetown University is a diverse community, the Admissions Committee would like to know more about you in your words. Please submit a brief essay, either personal or creative, which you feel best describes you.”* The demographic information of the authors can be found in Table S7. While the raw essay text cannot be shared due to confidentiality, all the processed data needed to replicate our analyses are available at OSF at <https://osf.io/ywap5>.

GPT-4 Essays

For AI-generated essays, we used GPT-4 to produce college admissions essays of approximately 650 words, matching the average length of human-authored essays. Specifically, we fed the same prompt provided to human applicants to the GPT-4 model, but we added the following instruction ahead of the original essay prompt: *“As a college applicant, please generate my admissions essay of approximately 650 words in response to the following prompt: [Essay Prompt].”* All GPT-4 essays generated in Studies 1 to 3, along with the personal backgrounds utilized for Chain-of-Thought prompting in Study 3, are available on OSF at <https://osf.io/ywap5>.

Across studies, essay lengths were comparable across human- and GPT-4–generated texts (see the Supplementary Online Material [SOM] for details at <https://osf.io/uvq2b>).

Measures

Semantic Distance: Divergent Semantic Integration

We quantified the creativity level of each essay using a semantic distance method: Divergent Semantic Integration (DSI; Johnson et al., 2022). Various empirical studies have shown that computational measures of semantic distance are positively associated with human ratings of creativity, helping to overcome the challenges of subjectivity and labor intensiveness in human evaluations (Beaty & Johnson, 2021; Heinen & Johnson, 2018; Johnson et al., 2021, 2022). Orwig et al. (2024) reported that the semantic distance-based measure was strongly correlated with creativity evaluations by human raters, both for human-written short stories ($r = 0.56$) and for stories generated by the GPT-3 ($r = 0.56$) and GPT-4 ($r = 0.44$).

The DSI measures creativity by computing the average distance between all word pairs in the content. The distance between two words is determined using the cosine distance between their respective word embeddings. Word embeddings are high-dimensional numerical vectors that represent words based on their distribution or co-occurrence with other words across a large corpus. They capture the principle of distributional semantics, which suggests that words occurring in the same contexts tend to have similar meanings (Harris, 1954). While the numerical values of word embeddings themselves may not have symbolic meaning, they can effectively measure semantic similarity between words and texts. A larger average distance between words indicates the presence of diverse concepts and ideas, thus indicating greater creativity. Conversely, a smaller average distance suggests a narrower range of concepts and ideas, indicating lower creativity.

Among the various methods available for generating word embeddings, Google's Bidirectional Encoder Representations from Transformers (BERT; Devlin et al., 2019) outperforms older models such as the Latent Semantic Analysis (LSA; Landauer & Dumais, 1997), the continuous bag-of-words model (CBOW; Mikolov et al., 2013), or Global Vectors (GloVe; Pennington et al., 2014). Unlike these older models that produce context-independent embeddings, transformer-based models such as BERT

generate context-dependent word embeddings. This means that BERT is sensitive to the actual use of words in various contexts, addressing the limitations of earlier models that had identical representations of homonyms or struggled with polysemy (Klein & Murphy, 2001). For instance, BERT can differentiate between the word "lead" used in the context of "lead a team" and "made of lead". A primary feature that contributes to BERT's nuanced understanding of context is the self-attention mechanism. This allows each word in a sentence to weigh its influence on others, thereby tailoring the word representation according to the surrounding context. This attention to context, combined with BERT-large's 340 million parameters, enables BERT to capture the richness of contextual meaning derived from factors such as syntax, word order, and word choice (Clark et al., 2019). As a result, BERT and other transformer-based models outpace earlier models, showing superior performance in downstream tasks (Castro et al., 2024; Devlin et al., 2019).

In the case of college admissions essays, semantic distance can capture both the diversity of past experiences and the creativity with which people express and integrate those experiences into a coherent narrative. For example, an applicant who participated in a high school robotics competition could describe the experience in a matter-of-fact way, such as *"In high school, I participated in a robotics competition where we built and programmed robots to complete specific tasks."* This student could be more creative in describing the same experience by broadening the context and introducing a more complex and abstract connection between diverse concepts, thus increasing the semantic distance: *"In high school, I participated in a robotics competition where we engineered robots that mimicked the neural connectome of *Caenorhabditis elegans* to create a self-navigating object."* Of course, semantic distance also additionally captures the diversity, uniqueness, and novelty of people's past experiences. For example, a student who describes her fairly common volunteer experience in a soup kitchen will likely have a narrower semantic distance than a student who describes a fairly uncommon volunteer experience, such as participating in drone surveillance for orangutan conservation in Borneo's lowland

forests. In short, semantic distance allows us to capture whether the GPT model may have a homogenizing effect on the collective diversity of ideas and experiences that the model expresses when mimicking human-generated text.

Finally, although we used the word-level embeddings following the DSI method as an established approach for measuring the creative divergence (Johnson et al., 2022), we also conducted a robustness check using sentence-level embeddings from the Sentence-BERT, specifically the all-mpnet-base-v2 model, which is recognized for strong performance on semantic similarity tasks (Reimers & Gurevych, 2019). Because sentence-level embeddings were computationally more efficient, they did not require the more extensive sampling procedures described in the following sections. Consequently, we tested both approaches, with and without using sampling procedures used in the word-level embeddings, to ensure robustness. All of these alternative approaches produced highly comparable results, indicating that our main results are robust to variations in embedding methods and sampling constraints (see Figures S1–S3 and Tables S6–S7).

Individual Diversity

Following the steps outlined in previous research (Johnson et al., 2022), we computed the DSI for each essay based on word embeddings to measure individual diversity. We tokenized all essays into WordPiece units using the BERT tokenizer and embedded them with the bert-large-uncased model (Devlin et al., 2019). Because the uncased model was used, all text was automatically lowercased during preprocessing. Stopwords were retained to preserve contextual meaning, as BERT embeddings are context sensitive. However, special tokens and selected punctuation ([CLS], [PAD], [SEP], ., ,, !, ?) were excluded from embedding calculations. Sentence segmentation was performed with NLTK's PunktSentenceTokenizer (Kiss & Strunk, 2006). Then, we computed the cosine distances for all possible combinations of embeddings extracted from layers 6 and 7 of the BERT model, following prior validation work (Johnson et al., 2022). Finally, we averaged the cosine distances to obtain the DSI score, which

reflects the semantic distance of the words used in each essay. Johnson et al. (2022) showed that higher DSI scores represent the presence of diverse concepts and ideas, indicating greater creativity. In contrast, lower DSI scores indicate a more limited range of concepts, suggesting lower creativity. In a previous study, DSI scores accounted for up to 72% of the variance in human creativity evaluations of creative narratives (Johnson et al., 2022).

Because the original DSI scores varied on a very small scale (e.g., SDs < 0.008), we multiplied them by 1,000 to enhance readability in all subsequent analyses, following a similar approach used in Doshi and Hauser (2024). This rescaling allows values to be reported with greater precision (e.g., two decimal places). Notably, this scaling adjustment does not influence the statistical inferences.

While the average individual diversity among human-written essays remained consistent across all three studies, the individual diversity among the base GPT-4 essays showed greater variation. Specifically, we observed the lowest average individual diversity in the base GPT-4 essays in Study 3, which utilized a more up-to-date GPT-4 model. This variation may reflect differences in the behavior of the various GPT-4 models used for essay generation, despite the use of the same prompt and settings across the studies.

Diversity Growth Rate of Collective Diversity

To quantify LLM-driven homogenization, we developed a new measure: the diversity growth rate. This measure shows how much each new essay adds new ideas beyond those already covered. The first few essays usually bring many new ideas, but later essays add progressively fewer. If there was a stronger homogenizing effect, each additional essay would lead to a smaller increase in collective diversity, resulting in a lower diversity growth rate.

To compute the diversity growth rate, we first obtained a measure of collective diversity, which captures the overall semantic diversity in a group of essays. We operationally defined collective diversity as the average semantic distance between word pairs within a collection of multiple essays. College

admissions essays, typically a few hundred words long, are much longer than the texts usually used in previous research (Cox et al., 2023; Meincke et al., 2024), which are often just a few sentences. This length introduced computational challenges, as calculating semantic distances between all word pairs in a group of multiple essays requires exponentially more computations. To manage this, we developed a novel approach involving sentence sampling and bootstrapping. We describe this method in four steps, using a group of two human-written essays as examples.

In Step 1, we randomly selected and combined a pair of essays drawn from the human-written essays in our sample. In Step 2, we randomly selected 34 sentences¹ from this pooled essay, the average length of the individual essays. Thus, we maintained consistency in length when calculating collective diversity, irrespective of the number of essays selected. In Step 3, we computed the DSI scores of the 34 selected sentences following the same procedure used for computing individual diversity. In Step 4, we repeated Steps 1 through 3 over 5,000 times; by averaging the resulting 5,000 DSIs, we obtained a robust measure of semantic diversity inherent in two human-written essays. We repeated this procedure by aggregating three human-written essays and so forth, up to a maximum of 34 essays.² We repeated the same procedure to assess the collective diversity of GPT-4 essays. As a result, we obtained 33 scores of collective diversity for each essay group, with the number of essays ranging from 2 to 34.

Using the collective diversity scores from 2 to 34 essays, we were able to compute the diversity growth rate—the magnitude of the changes in collective diversity as the number of selected essays increased.

Analytic Strategy

¹ Since the BERT model can capture the contextual meaning of words within a sentence, we chose to sample sentences rather than sample words.

² We capped the number of essays at 34 because increasing the group size beyond this point did not enhance collective diversity. Specifically, when creating an aggregated essay, 34 sentences are selected from the chosen essays; thus, any additional essays beyond 34 cannot contribute sentences, rendering them ineffective at enhancing diversity.

To test our hypotheses regarding individual diversity, we compared the mean DSI scores between different authorship groups. In Study 1, which involved a single comparison, we used Welch's t -test to account for unequal variances between the two groups. In Studies 2 and 3, which involved multiple group comparisons, we first conducted an omnibus ANOVA to test for overall effects. We then conducted planned pairwise post-hoc comparisons, adjusting p -values for multiple comparisons using the Holm correction.

To test our hypotheses regarding diversity growth rates, we used a two-step approach centered on multiple regression analysis. First, we predicted collective diversity from the natural log-transformed number of essays, dummy-coded variables for essay authorship (with human-written essays as the reference category), and their interactions. We applied the logarithmic transformation because increases in collective diversity follow a pattern of diminishing returns: the first few essays in a group typically add many novel ideas, but each additional essay contributes progressively fewer. The log curve captures this tapering growth, similar to rarefaction curves in ecology, where sampling more organisms increases observed species diversity but at a slowing rate.

From this model, we estimated the specific diversity growth rate for each authorship condition, defined as the slope (b) of the log-transformed number of essays. In plain terms, this slope represents the expected gain in collective diversity as group size expands. Smaller slopes therefore indicate stronger homogenization, as each new essay contributes less additional diversity to the group. Technically, the slope (b) indicates that a 1% increase in the number of essays is associated with a $b/100$ unit increase in collective diversity; we reported the effect of doubling the number of essays

(100% increase), calculated by multiplying b by $\ln(2)$.⁴ Second, we conducted post-hoc pairwise comparisons to test for significant differences between these estimated slopes, with p -values adjusted for multiple comparisons using the Holm correction.

Study 1

In Study 1, we conducted a baseline comparison between human-written and GPT-4 generated college admissions essays. We hypothesized that human-written essays would show a higher level of individual diversity compared to base GPT-4 essays. We also expected that human-written essays would show a higher diversity growth rate compared to GPT-4 essays.

Methods

We compared a set of 100 essays written by humans randomly chosen from submissions between 2018 and 2022 with another set of 100 essays generated using GPT-4. Since the Application Programming Interfaces (APIs) for OpenAI's GPT-4 model were not accessible to us during the data generation phase of Study 1, we manually produced 100 essays by using ChatGPT through its chat interface between June 24th and 30th, 2023. Each essay was generated in a new chat session. While GPT-4 essays often included titles or footers, we excluded these from our analyses since the human-authored essays did not contain titles or footers.

Results

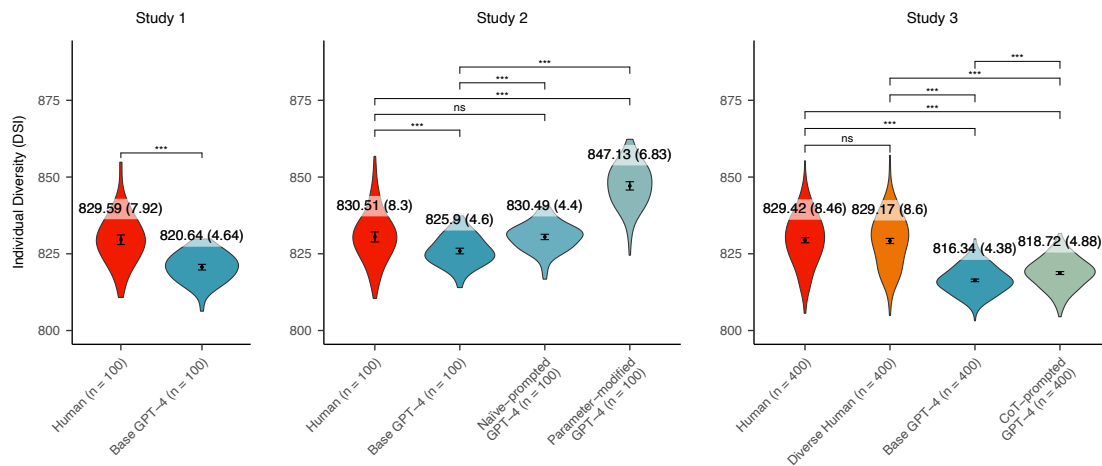
Individual Diversity

⁴ In each study, we also ran multiple regression models that included only the log-transformed number of essays and dummy-coded essay authorship. This allowed us to estimate the mean differences between authorship groups while holding the number of essays constant. It also allowed us to examine the effect of the log-transformed number of essays on increasing collective diversity, on average, across different authorship groups. Across all three studies, we found that the log-transformed number of essays in a group was positively associated with collective diversity. As the number of essays in a group increased, the group's collective diversity also increased. See Table S1 in the SOM for details.

To examine whether human-written essays would show a higher level of individual diversity compared to base GPT-4 essays, we compared the average DSI scores of individual essays. As expected, the average human-written essay was more semantically diverse compared to the base GPT-4 essays, $Mean\ Difference = 8.95$, $t(159.89) = 9.75$, $p < .001$, Cohen's $d = 1.38$ (Figure 1 and Table 1).

Figure 1

Comparison of Individual Diversity Scores (DSI) in Human And GPT-4 Essays Across Studies



Note. The distributions and differences in individual diversity scores across authorship conditions. The error bars represent the 95% confidence intervals. *** $p < .001$.

Table 1

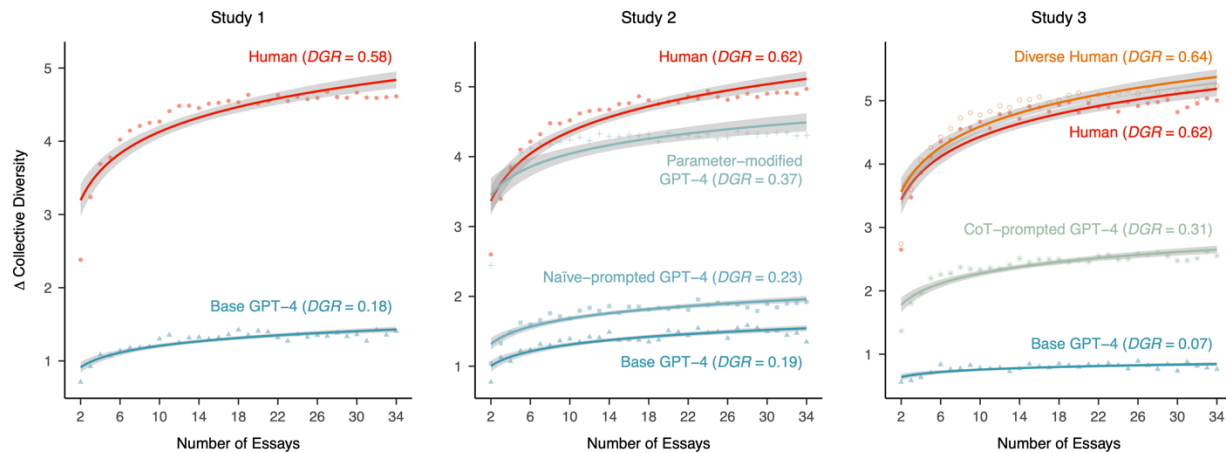
Planned-Pairwise Comparisons of Individual Diversity Scores (DSI) by Authorship Conditions Across Studies

Study	Group 1	Group 2	Mean Difference	95% CI	Cohen's <i>d</i>	<i>t</i>	<i>p</i> / <i>p</i> _{holm}
Study 1	Human	Base GPT-4	8.95	[7.14, 10.76]	1.38	9.75	<.001
Study 2	Human	Base GPT-4	4.61	[2.87, 6.34]	0.74	5.21	<.001
Study 2	Naïve-prompted GPT-4	Base GPT-4	4.59	[2.85, 6.32]	0.73	5.19	<.001
Study 2	Parameter-modified GPT-4	Base GPT-4	21.22	[19.49, 22.96]	3.40	24.02	<.001
Study 2	Human	Naïve-prompted GPT-4	0.02	[-1.72, 1.76]	0.00	0.02	.982
Study 2	Human	Parameter-modified GPT-4	-16.62	[-18.35, -14.88]	-2.66	-18.81	<.001
Study 3	Human	Base GPT-4	13.08	[12.13, 14.03]	2.09	26.94	<.001
Study 3	CoT-prompted GPT-4	Base GPT-4	2.38	[-1.43, 3.33]	0.35	4.90	<.001
Study 3	CoT-prompted GPT-4	Human	-10.70	[-11.65, -9.75]	-1.56	-22.04	<.001
Study 3	Diverse human	Human	-0.25	[-1.20, 0.71]	-0.04	-0.51	0.613
Study 3	Diverse Human	Base GPT-4	12.83	[11.88, 13.78]	1.87	26.43	<.001
Study 3	Diverse Human	CoT-prompted GPT-4	10.45	[9.50, 11.41]	1.52	21.53	<.001

Note. We found a significant difference in average DSI scores by authorship conditions, in Study 2, $F(3, 396) = 223.23$, $p < .001$, $\eta^2 = .63$; and in Study 3, $F(3, 1596) = 399.70$, $p < .001$, $\eta^2 = .43$. *P*-values in Studies 2 and 3 were adjusted for multiple comparisons using the Holm correction. No correction was applied in Study 1 because there was only one comparison.

Diversity Growth Rate

The estimated diversity growth rate and doubling effect for each condition are presented in Table 2. The comparisons of these growth rates between authorship conditions are illustrated in Figure 2 and detailed in Table S3. As expected, the human-written essays showed a higher diversity growth rate compared to the GPT-4 essays, suggesting a more pronounced homogenizing effect in GPT-4 essays. Each doubling in the number of human-written essays led to a 0.40 increase in collective diversity, $b = 0.58$, 95% CI = [0.50, 0.66], $p < .001$. In contrast, each doubling in the number of base GPT-4 essays only led to a 0.13 increase, $b = 0.18$, 95% CI = [0.11, 0.26], $p < .001$. This corresponds to only about 31% of the diversity growth rate observed for human-written essays, $b = 0.40$, 95% CI = [0.29, 0.50], $p < .001$.

Figure 2*Comparison of Diversity Growth Rates in Human and GPT-4 Essays Across Studies*

Note. The figures illustrate changes in collective diversity with the increasing number of essays across different essay authorships. Here, we plot the difference between collective and individual diversity (collective minus individual), rather than raw collective diversity, to show the net benefit of group-level contributions beyond what is already present at the individual level. The Y-axis therefore reflects these net diversity benefits of adding additional essays to a group. Individual diversity was measured by the average semantic distance between all word pairs within each essay, while collective diversity was measured by the average semantic distance between word pairs within a collection of multiple essays. The fitted curved lines represent the diversity growth rate (DGR) for each type of essay, with all slopes being significant at $p < .001$, except for the base GPT-4 condition in Study 3 ($p = 0.071$).

Table 2*Diversity Growth Rates (DGR) by Authorship Conditions*

Study	Authorship	DGR (<i>b</i>)	95% CI	Doubling Effect	SE	df	<i>t</i>	<i>p</i>
S1	Human	0.58	[0.50, 0.66]	0.40	0.04	62	15.35	< 0.001
S1	Base GPT-4	0.18	[0.11, 0.26]	0.13	0.04	62	4.82	< 0.001
S2	Human	0.62	[0.54, 0.70]	0.43	0.04	124	15.67	< 0.001
S2	Base GPT-4	0.19	[0.11, 0.27]	0.13	0.04	124	4.84	< 0.001
S2	Naïve-prompted GPT-4	0.23	[0.15, 0.30]	0.16	0.04	124	5.75	< 0.001
S2	Parameter-modified GPT-4	0.37	[0.29, 0.45]	0.25	0.04	124	9.33	< 0.001
S3	Human	0.62	[0.54, 0.70]	0.43	0.04	124	15.50	< 0.001
S3	Diverse Human	0.64	[0.56, 0.72]	0.44	0.04	124	16.09	< 0.001
S3	Base GPT-4	0.07	[-0.01, 0.15]	0.05	0.04	124	1.82	0.071
S3	CoT-prompted GPT-4	0.31	[0.23, 0.39]	0.21	0.04	124	7.75	< 0.001

Note. The Diversity Growth Rate (DGR) represents the estimated slope (*b*) for each authorship condition.

The Doubling Effect, calculated as $DGR * \ln(2)$, shows the estimated increase in collective diversity from doubling the number of essays. 95% CI = 95% Confidence Interval; SE = Standard Error.

Study 2

In Study 1, we found that human-written essays showed not only higher individual diversity but also a higher diversity growth rate, compared to base GPT-4 essays. In Study 2, we aimed to replicate the baseline findings of Study 1 while examining whether the homogenizing effect of GPT-4 persists after enhancing its output. We enhanced the idea diversity of GPT-4-generated essays through naïve prompt engineering and parameter modifications. We analyzed 400 essays, divided equally into four groups of 100 each: (1) human-written essays, (2) base GPT-4 essays, (3) naïve-prompted GPT-4 essays, and (4) parameter-modified GPT-4 essays. We hypothesized that human-written essays would show a higher level of individual diversity compared to base and naïve-prompted GPT-4 essays, but not as much as parameter-modified GPT-4 essays. We also expected that human-written essays would still show a higher diversity growth rate compared to all three types of GPT-4 essays.

Method

We used OpenAI's APIs to generate three different versions using the 'gpt-4-0613' model between July 19 and July 24, 2023. First, as in Study 1, we generated the base GPT-4 essay. Second, we employed naïve prompt engineering by adding the phrase "be as creative as possible in your writing" at the end of the essay generation prompt. Finally, we modified two parameters of the GPT-4 model, 'presence_penalty' and 'frequency_penalty', from the default value of 0 to the maximum value of 2. The 'presence_penalty' parameter controls the model's tendency to use new words that it has not used before, while the 'frequency_penalty' parameter guides the model to avoid using frequently occurring words. By increasing these parameters, we expected the models to generate more diverse and creative text, potentially leading to less repetition and more novel outputs.⁵

⁵ The "temperature" parameter can also influence the diversity of text generation outputs; however, we chose not to modify this parameter. In our pilot study, adopting a maximum temperature value of 2, instead of modifying presence_penalty or frequency_penalty, more often resulted in nonsensical prose that was hard to understand or failed to complete text generation.

Results

Individual Diversity

We found the significant omnibus effect of essay authorship on individual diversity, $F(3, 396) = 223.23$, $p < .001$, $\eta^2 = .63$, indicating that about 63% of the variance in individual diversity was explained by essay authorship. The manipulation check confirmed that adopting either the naïve or parameter modifications increased individual diversity of GPT-4 essays. Specifically, compared to the base GPT-4 essays, the naïve-prompting led to an increase of 4.59, $t(396) = 5.19$, $p_{\text{holm}} < .001$, Cohen's $d = 0.73$; and the parameters-modification led to an increase of 21.22, $t(396) = 24.02$, $p_{\text{holm}} < .001$, Cohen's $d = 3.40$.

Replicating the results in Study 1, the human-written essays showed higher individual diversity compared to the base GPT-4 essays, *Mean Difference* = 4.61, $t(396) = 5.21$, $p_{\text{holm}} < .001$, Cohen's $d = 0.74$ (Table 1). Importantly, as expected, the parameter-modified model exceeded the individual diversity of the human-written essays, *Mean Difference* = 16.62, $t(396) = 18.81$, $p_{\text{holm}} < .001$, Cohen's $d = 2.66$. However, contrary to our preregistered prediction, there was no significant difference in individual diversity between the human-written essays and the naïve-prompted GPT-4 essays, *Mean Difference* = 0.02, $t(396) = 0.02$, $p_{\text{holm}} = .982$, Cohen's $d = 0.00$.

Diversity Growth Rate

The estimated diversity growth rate and doubling effect for each condition are presented in Table 2. The comparisons of these growth rates between authorship conditions are illustrated in Figure 2 and detailed in Table S3. Replicating the results in Study 1, we found that the diversity growth rate for the base GPT-4 essays was only about 31% of the rate for the human-written essays, $b = 0.43$, 95% CI = [0.28, 0.58], $p_{\text{holm}} < .001$. Specifically, each doubling of the number of human-written essays led to a 0.43 increase in collective diversity, $b = 0.62$, 95% CI = [0.54, 0.70], $p < .001$. In contrast, the increase for the base GPT-4 essays was only 0.13, $b = 0.19$, 95% CI = [0.11, 0.27], $p < .001$.

Next, as a manipulation check, we tested our assumption that adopting naïve prompting or parameter modifications would reduce the homogenizing effect in GPT-4 outputs. The naïve-prompted GPT-4 essays showed no significant difference in diversity growth rate compared to the base GPT-4 essays, $b = -0.04$, 95% CI = $[-0.19, 0.11]$, $p_{\text{holm}} = .520$. However, parameter modifications led to a significant increase in the diversity growth rate, approximately 93% higher than that of GPT-4 essays, $b = -0.18$, 95% CI = $[-0.33, -0.03]$, $p_{\text{holm}} = .006$. Specifically, each doubling of the number of the parameter-modified GPT-4 essays led to a 0.25 increase in collective diversity, $b = 0.37$, 95% CI = $[0.29, 0.45]$, $p < .001$. As expected, despite this improvement, the human-written essays still showed a significantly higher diversity growth rate compared to both the prompt- and parameter-modified GPT-4 essays. Specifically, the diversity growth rate of the human-written essays was about 170% higher than that of the naïve-prompted GPT-4 essays, $b = 0.39$, 95% CI = $[0.24, 0.54]$, $p_{\text{holm}} < .001$, and about 67% higher than that of the parameter-modified GPT-4 essays, $b = 0.25$, 95% CI = $[0.10, 0.40]$, $p_{\text{holm}} < .001$. In short, we found that human-written essays showed a higher diversity growth rate compared to all three types of GPT-4 essays.

Study 3

Study 2 replicated the baseline pattern that human-written essays showed not only higher individual diversity but also a greater diversity growth rate compared to base GPT-4 essays. Notably, while the parameter modifications increased the individual diversity of GPT-4 essays to levels matching or even exceeding that of human-written essays, human-written essays still showed a greater diversity growth rate. These findings highlight the persistent homogenizing effect in creative LLM outputs, despite efforts to enhance semantic diversity of individual LLM outputs.

In Study 3, we had three primary objectives. First, we aimed to replicate our findings from Studies 1 and 2 with a larger sample ($n = 1,600$). Second, we tested the effect of Chain-of-Thought (CoT) prompting on individual diversity and diversity growth rate of GPT-4 essays as an alternative to

parameter modifications. While parameter modifications increased semantic diversity of individual GPT-4 essays by introducing more randomness, they often led to overly nonsensical outputs. To address this, we employed CoT prompting because it is considered a state-of-the-art approach for enhancing the performance of LLMs by enabling step-by-step reasoning that mimics the human cognitive processes (Wei et al., 2023). In Studies 1 and 2, we explored how prompt and parameter modifications affected both the individual and collective diversity of GPT-4 essays. However, factors that might influence the diversity of human work were left unexplored. Therefore, our third goal was to examine whether diversifying the racial and ethnic backgrounds of human authors would promote the diversity growth rate.

We analyzed 1,600 essays, divided equally into four groups: (1) human-written essays from random applicants, (2) human-written essays from racially and ethnically diverse applicants, (3) base GPT-4 essays, and (4) CoT-prompt GPT-4. We hypothesized that human-written essays from both random applicants and applicants with racially and ethnically diverse backgrounds would show a higher level of individual diversity compared to both base and CoT-prompted GPT-4 essays. We also expected higher diversity growth rates in human-written essays from both applicant groups compared to the two types of the GPT-4 essays, with an even greater increase for essays from racially and ethnically diverse applicants.

Method

We used OpenAI's APIs to generate two sets of the GPT-4 essays: (1) base GPT-4 essays and (2) Chain-of-Thought (CoT) prompted GPT-4 essays. Initially, as in Studies 1 and 2, we generated base GPT-4 essays, but this time, we produced a larger quantity of 400 essays using the newer version of the 'gpt-4-0125-preview' model on March 27, 2024. Then, we generated another set of 400 GPT-4 essays on May 28, 2024, using a sophisticated prompt engineering technique known as Chain-of-Thought (CoT) prompting (Meincke et al., 2024). This process involved the following two steps: First, we generated

personal backgrounds for 400 hypothetical college applicants using GPT-4. Second, we utilized CoT prompting to generate admissions essays based on these backgrounds. We expected that CoT prompting would guide the GPT-4 model to reason through a task step-by-step, enhancing the diversity of the essays (for further methodological details, refer to the Python code for essay generation available at OSF <https://osf.io/ywap5>).

Results

Individual Diversity

To examine differences in individual diversity depending on essay authorship, we compared the average DSI scores of individual essays across essay groups. We found the significant omnibus effect of essay authorship on individual diversity, $F(3, 1596) = 399.70, p < .001, \eta^2 = .43$, indicating that about 43% of the variance in individual diversity was explained by essay authorship. Manipulation check confirmed that adopting CoT prompting increased individual diversity of the GPT-4 essays by 2.38, $t(1596) = 4.90, p_{\text{holm}} < .001$, Cohen's $d = 0.35$.

Replicating the results of Studies 1 and 2, the human-written essays, particularly those from the random applicant group, showed significantly higher individual diversity compared to the base GPT-4 essays, *Mean Difference* = 13.08, $t(1596) = 26.94, p_{\text{holm}} < .001$, Cohen's $d = 2.09$. Moreover, despite improvements in the CoT-prompted GPT-4 essays, the human-written essays from the random applicant group still exhibited significantly higher individual diversity compared to the CoT-prompted GPT-4, *Mean Difference* = 10.70, $t(1596) = 22.04, p_{\text{holm}} < .001$, Cohen's $d = 1.56$. Similarly, the human-written essays from the racially and ethnically diverse human group also showed significantly higher individual diversity compared to the base GPT-4 essays, *Mean Difference* = 12.83, $t(1596) = 26.43, p_{\text{holm}} < .001$, Cohen's $d = 1.87$; and to the CoT-prompted GPT-4 essays, *Mean Difference* = 10.45, $t(1596) = 21.53, p_{\text{holm}} < .001$, Cohen's $d = 1.52$. There was no significant difference in individual diversity between the random

applicant group and the racially and ethnically diverse applicant group, *Mean Difference* = 0.25, $t(1596) = 0.51$, $p_{\text{holm}} = .613$, Cohen's $d = 0.04$.

Diversity Growth Rate

The estimated diversity growth rate and doubling effect for each condition are presented in Table 2. The comparisons of these growth rates between authorship conditions are illustrated in Figure 2 and detailed in Table S3. Contrary to our preregistered predictions, there was no difference in the diversity growth rate between human-written essays from the random applicant group and those from the racially and ethnically diverse applicant group, $b = -0.02$, 95% CI = [-0.17, 0.13], $p_{\text{holm}} = .680$. Thus, for more direct comparisons with Studies 1 and 2, we focus on comparing the human-written essays from the random applicant group to the GPT-4 generated essays.

Replicating the results of Studies 1 and 2, we found that the diversity growth rate for the base GPT-4 essays was only about 11% of the rate for the human-written essays, $b = 0.54$, 95% CI = [0.39, 0.69], $p_{\text{holm}} < .001$. Specifically, each doubling of the number of the randomly selected human-written essays led to a 0.43 increase in collective diversity, $b = 0.62$, 95% CI = [0.54, 0.70], $p < .001$. In contrast, the increase for the base GPT-4 essays was only 0.05 and not statistically significant, $b = 0.07$, 95% CI = [-0.01, 0.15], $p = .071$.

Next, we tested our assumption that adopting CoT prompting would reduce the homogenizing effect in GPT-4 outputs. We found that CoT prompting increased the diversity growth rate more than four times higher than that of the base GPT-4 essays, $b = 0.24$, $t(124) = 4.19$, $p < .001$. Specifically, each doubling of the number of the CoT-prompted essays led to a 0.21 increase in collective diversity, $b = 0.31$, 95% CI = [0.23, 0.39], $p < .001$. Despite these improvements, human-written essays still showed about twice as high diversity growth rate compared to the CoT-prompted GPT-4 essays, $b = 0.31$, 95% CI = [0.16, 0.46], $p < .001$. Overall, we found higher diversity growth rates in human-written essays than in both of the GPT-4 conditions.

Discussion

Across three preregistered studies, we assessed 2,200 college admissions essays to compare the creative capabilities of humans and GPT-4, focusing on the semantic diversity of creative writings. Using a novel measure of the diversity growth rate, we showed that each additional human-written essay contributed more new ideas than each additional GPT-4 essay. In three studies, human writing increased the collective semantic diversity of a group of essays approximately two to eight times more than base GPT-4 essays did. This homogenizing effect persisted even after a range of enhancements to the diversity of the GPT-4 writings, including prompt and parameter modifications. Even when these modifications led to more semantically diverse essays than human-written essays at the individual level, GPT-4 had a greater homogenizing effect than human writers. Thus, even as AI tools continue to improve and produce ever better and more creative output, they may still contribute to an overall homogenization of ideas. In short, our findings suggest that the widespread use of LLMs could diminish the diversity of ideas at the collective level.

Our research addresses critical gaps in the literature on the creative capabilities of LLMs by introducing a dynamic perspective on the homogenizing effect in LLMs through our novel metric: the diversity growth rate. Previous studies have found that GPT-4 outputs are often more homogenized than human-generated content, such as in product ideas (Meincke et al., 2024), motivational messages (Cox et al., 2023), admissions essays (Alvero et al., 2022), or even in co-creations produced by hybrid writing between humans and LLMs (Anderson et al., 2024; Padmakumar & He, 2024). However, these studies primarily adopted static approaches, which involved examining homogenization in LLMs by comparing the collective diversity of content sets with a fixed quantity of outputs. As a result, how homogenization changes with an expanding volume of content has remained unexplored until now.

By shifting from a static to a dynamic view, our study provides a better understanding of how the difference in contribution to collective diversity between human- and AI-generated content grows as

more content is considered. For example, as shown in our data, a small sample of GPT-4 essays might initially appear somewhat varied; but as the volume of essays grows, it becomes increasingly evident that GPT-4 essays contribute significantly fewer novel ideas compared to essays written by human writers. This result underscores the importance of exploring not just whether AI-generated outputs are less diverse, but also how quickly and to what extent this reduced diversity accumulates when large quantities of AI-generated work are considered together.

From a theoretical standpoint, our findings highlight the potential population-level ramifications of widespread LLM adoption. If organizations, educational institutions, or creative industries rely too much on a certain AI model, the collective pool of ideas may become more uniform over time. AI can thus inhibit innovation and diminish cultural diversity. This homogenization is especially counterproductive in fields that thrive on diverse perspectives—such as technology, healthcare, and education (Kaufman & Sternberg, 2019)—and underscores the importance of “thinking outside the AI box,” particularly when tackling the toughest challenges. LLMs, like any other generative AI models, are fundamentally based on existing data and previously explored ideas. The novel diversity growth curve can be used to develop effective mitigation strategies by understanding when and at what saturation point this homogenizing effect accelerates or plateaus. Indeed, the diversity growth rate across AI-generated content offers a systematic approach to monitoring and mitigating the risk of LLM-driven homogenization. By identifying when and how much the introduction of novel ideas begins to slow or converge, practitioners can identify high-risk saturation points and intervene early—whether by adjusting model parameters, diversifying training data, advancing prompting strategies, or augmenting human oversight—to preserve creative variety.

One might expect the tendency of LLMs toward homogenization to decrease as GPT models and other LLMs continue to evolve. However, previous research suggests that LLMs share two systemic factors that inherently limit their output diversity. First, these models are autoregressive systems

optimized to select the most probable next token, which fundamentally favors predictable and conventional outputs over novel or unexpected ones (Brown et al., 2020). Second, they undergo instruction tuning with Reinforcement Learning from Human Feedback (RLHF) to produce “safe” or aligned responses (Kirk et al., 2024; Mohammadi, 2024). These processes, while improving fidelity and reducing hallucinations, can inadvertently reduce creativity and encourage homogenized responses. Indeed, our findings showed that a newer version of GPT-4 (gpt-4-0125-preview) actually yielded higher homogenization than the previous model (gpt-4-0613). Further supporting this tendency, adjusting parameters such as ‘presence_penalty’ and ‘frequency_penalty’ increased semantic diversity and decreased homogenization but at the expense of coherence and readability (see Table S10 for details). Even with these modifications, AI-generated essays still had a lower diversity growth rate than those written by humans, suggesting that homogenization may be an inherent feature of current LLMs. Nevertheless, future work should not only continue quantifying the homogenizing effects of new GPT versions and other LLMs (e.g., Meta’s Llama, Claude Sonnet, Google’s Gemini), but also use the diversity growth rate to optimize the model to reduce homogenization.

Our research has several notable limitations. First, the computational measure of semantic diversity in this study primarily captures creative divergence, which focuses on the generation of novel ideas (Acar & Runco, 2019; Johnson et al., 2022), but does not account for convergent thinking—such as logical coherence, problem-solving efficiency, and practical feasibility. To examine the creative capabilities of LLMs in convergent thinking, a recent study used a novel dataset of real-world problem-solving tasks; the results showed that while LLMs could generate diverse solutions, they often propose impractical or physically infeasible responses (Tian et al., 2024). Similarly, another study evaluated LLMs’ abstract reasoning using the New York Times *Connections* word game, revealing that models struggled with multi-word expressions and encyclopedic knowledge compared to human players (Samdarshi et al., 2024). These findings suggest that while LLMs may exhibit surface-level diversity, their collective

contribution to meaningful and logically sound creativity remains limited. Future research should explore how LLMs perform in tasks requiring collective convergent thinking, such as assessing the logical coherence and contextual appropriateness, while simultaneously preserving diversity in their responses. A deeper investigation using the diversity growth approach outlined here could provide a more comprehensive understanding of the homogenizing effects of LLMs.

Additionally, we aimed to measure how much humans versus GPT-4 differently contribute to collective diversity, rather than examining homogenization in human-AI co-creation. This design allowed us to directly compare contributions to group-level diversity by examining essays written by multiple humans against multiple outputs generated by the same LLM. However, this comparison does not fully reflect how LLMs are typically used in practice, where individuals may provide personal background information and iteratively refine model's outputs. In our base condition, GPT-4 was asked to "write as a college applicant" without individualized context, which may have encouraged more generic responses. In Study 3, we attempted to introduce personalization by generating applicant backgrounds with GPT-4, but this procedure is still limited compared to the personal information LLM-based chat agents have about each user in practice. In this sense, our findings illuminate the homogenizing effects of LLMs in controlled conditions, while calling for future research to examine whether similar patterns hold in more naturalistic, co-creative settings.

Finally, the generalizability of our findings is also limited. Our analyses involved only essays written by college applicants and were conducted in the specific context of writing college admissions essays. This limits the extent to which our findings can be applied to other populations and different writing contexts. Additionally, the limited diversity among human writers in our study may explain the null findings in Study 3 regarding the lack of significant differences in collective diversity between essays written by random applicants and diverse applicants. Indeed, our participants were already homogeneous in age, educational level, and the context of applying to the same university. This

suggests, however, that our findings may underestimate the homogenizing effect of LLMs compared to a group of more diverse individuals.

Despite these limitations, the present research provides a proof of concept and blueprint for investigating how AI may homogenize ideas over time. To better understand LLM-driven homogenization, future studies should explore diverse writing domains such as academic papers, news articles, corporate reports, and social media. As more people use AI to assist with their writing, generate new ideas, and conduct literature reviews, the diversity growth rate can be used to quantify any homogenizing effect over time. Examining scientific publications, for example, could reveal formulaic trends in AI-generated abstracts, while news content may show stylistic convergence and fewer critical perspectives. Likewise, studying creative writing in literature or screenplays could shed light on how the increasing use of LLMs can impact originality. Broadening research across these areas will provide deeper insights into the collective effects of widespread use of AI on the diversity of ideas.

In conclusion, GPT-4 has limitations in producing semantically diverse content compared to groups of human authors. Using a novel metric, the diversity growth rate, we showed that while GPT-4 can generate creative content at the individual level, GPT-4 generated essays contribute less to the collective diversity of ideas, with a lower diversity growth rate than human-written essays. This homogenizing effect of GPT-4 remained despite a range of enhancements to the diversity of the GPT-4 writings, utilizing prompt and parameter modifications. Our results emphasize the need for AI literacy education and policies that encourage the diversity and originality of creative outputs from AI systems. Future research should explore homogenization in co-creative processes and extend analyses to various demographic and writing contexts to fully understand the impact of LLMs on creativity.

Data Availability

All materials to replicate the findings in this paper are available on a repository stored on the Open Science Framework (OSF) under a Creative Commons Attribution Non-Commercial 4.0

International (CC-BY-NC) license at <https://osf.io/ywap5>. The repository includes demographic data of human applicants, all raw GPT-4 generated essays, the code used to generate these essays, and the code and resulting data for computing individual and collective diversity scores used in this study. The raw text data from human-written essays cannot be shared due to confidentiality.

Code Availability

The code used for all analyses and generating plots in the main manuscript and SOM is available at <https://osf.io/ywap5>.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this work, the authors used GPT models in order to generate GPT-written essays and improve the language and readability of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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